7 Solutions

SOLUTION (QR CODE)

4. $\log\frac{P_{T_2}}{P_{T_1}} = \frac{\Delta H_{vap}}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ T = Kelvin

Substitute the values in the above expression as:

$$\log\frac{1}{P_{333}} = \frac{9800}{2.303 \times 2}\left[\frac{1}{333} - \frac{1}{373}\right]$$

On solving, P_{333} = 0.6851 atm

5. $\log\frac{P_{T_2}}{P_{T_1}} = \frac{\Delta H_{vap}}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ T = Kelvin

Substitute the values in the above expression:

$$\log\frac{2.5}{0.1} = \frac{\Delta H}{2.303R}\left[\frac{1}{1879} - \frac{1}{T}\right]$$

On solving we get T = 2169 K or 1896°C.

8. (B) P° = 20 torr, P_S = 17 torr

$P_S = P^\circ X_{solvent}$, 17 = 20 $X_{solvent}$

$X_{solvent} = \frac{17}{20} = 0.85$

9. In solution A ⇒ $X_{solvent}$ = 0.9

In solution B ⇒ 1 molal

1 mole चीनी in 1000 gm solvent

$$X_{solvent} = \frac{n_{solvent}}{n_{solvent} + n_{\text{चीनी}}} = \frac{\frac{1000}{18}}{\frac{1000}{18} + 1}$$

$X_{solvent}$ = 0.98

Solution B has more vapor pressure as compared to solution A.

10. Apply Raoult's law:

$$\frac{P_A^\circ - P_S}{P_A^\circ} = X_B = \frac{n_B}{n_B + n_A}$$

Calculate moles of solute as:

$$n_B = \frac{W_B}{M_B} = \frac{25}{180} = 0.1389$$

$$n_A = \frac{W_A}{M_A} = \frac{450}{18} = 25$$

Now, substitute the values in the above expression:

$$\frac{17.535 - P_S}{17.535} = \frac{0.1389}{0.1389 + 25}$$

P_S = 17.535 – 0.097 = 17.438 mm Hg

Question ki language se ghabra mat jana

12. $\frac{Y_A = \frac{n_A}{n_T}}{Y_B = \frac{n_B}{n_T}} = \frac{n_A}{n_B}$

$$X_A = \frac{1}{2};\ X_B = \frac{1}{2};\ P_P^\circ = 2P_Q^\circ$$

$$Y_A = \frac{P_A^\circ X_A}{P_A^\circ X_A + P_B^\circ X_B} = \frac{2P_B^\circ\left(\frac{1}{2}\right)}{2P_B^\circ\left(\frac{1}{2}\right) + P_B^\circ\left(\frac{1}{2}\right)} = \frac{P_B^\circ \times 2}{3P_B^\circ} = \frac{2}{3}$$

$$Y_B = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\frac{Y_A}{Y_B} = \frac{n_A}{n_B} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2:1$$

15. $P_T = P^\circ_A X_A + P^\circ_B X_B$

According to the given information,

$P^\circ_A \times 0.7 + P^\circ_B \times 0.3 = 350$ ...(i)

$P^\circ_A \times 0.2 + P^\circ_B \times 0.8 = 410$ ...(i)

From equations (i) and (ii):

P°_A = 314 mm Hg, P°_B = 434 mm Hg

18. $P_T = X_A P^\circ_A + X_B P^\circ_B$

Substitute the values in the above equation:

$= 0.6 \times 90 + 0.4 \times 15$

$= 54 + 6 = 60$

Now,

$X_A P^\circ_A = y_A P_T$, $0.6 \times 90 = y_A(60) \Rightarrow y_A = 0.9$

$y_B = 0.1 = 1 \times 10^{-1}$, $x = 1$

19. Mole fraction of A, $X_A = \frac{1}{4}$

Mole fraction of B, $X_B = \frac{3}{4}$

Calculate total vapor pressure as:

$= X_A.P^\circ_A + X_B.P^\circ_B$

$$550 = \frac{1}{4}P^\circ_A + \frac{3}{4}P^\circ_B$$

or $2200 = P^\circ_A + 3P^\circ_B$ ...(i)

After adding one more mole of B, the vapor pressure increases by 10 mm of Hg that is:

$$= \frac{1}{5}P^\circ_A + \frac{4}{5}P^\circ_B$$

$$560 = \frac{1}{5}P^\circ_A + \frac{4}{5}P^\circ_B$$

or $2800 = P^\circ_A + 4P^\circ_B$ ...(ii)

On solving equations (i) and (ii):

$P^\circ_B = 600$ mm of Hg, $P^\circ_A = 400$ mm of Hg

are asaan hai, mushkil lagra hai bas

21. $X_{A,1} = \frac{n_A}{n_A + n_B}$ in vessel 1

$$= \frac{3}{3+1} = \frac{3}{4}$$

$P^\circ_A = 700$ torr, $P^\circ_B = 350$ torr

(i) $$\frac{y_{A,1}}{1-y_{A,1}} = \frac{P^\circ_A}{P^\circ_B}\left(\frac{X_{A,1}}{1-X_{A,1}}\right)$$

$$\frac{y_{A,1}}{1-y_{A,1}} = \frac{700}{350}\left(\frac{\frac{3}{4}}{\frac{1}{4}}\right) = 2 \times 3 = 6$$

$y_{A,1} = 6 - 6y_{A,1}$

$$y_{A,1} = \frac{6}{7};\ y_{B,1} = \frac{1}{7}$$

(ii) $$\frac{y_{A,2}}{1-y_{A,2}} = \left(\frac{P^\circ_A}{P^\circ_B}\right)^2\left(\frac{X_{A,1}}{1-X_{A,1}}\right)$$

$$\frac{y_{A,2}}{1-y_{A,2}} = \left(\frac{700}{350}\right)^2\left(\frac{\frac{3}{4}}{\frac{1}{4}}\right) = 12$$

$$y_{A,2} = \frac{12}{13};\ y_{B,2} = \frac{1}{13}$$

(iii) $$\frac{y_{A,100}}{1-y_{A,100}} = \left(\frac{P^\circ_A}{P^\circ_B}\right)^{100}\frac{X_{A,1}}{1-X_{A,1}}$$

$$= \left(\frac{700}{350}\right)^{100}\left(\frac{\frac{3}{4}}{\frac{1}{4}}\right) = 3.2^{100} \simeq \infty$$

$y_{A,100} = 1$, $y_{B,100} = 0$

Dhyan lagao idhar

22. $P_T = P^\circ_A X_A + P^\circ_B X_B$

$$475 = P^\circ_A\left(\frac{\frac{100}{140}}{\frac{100}{140}+\frac{1000}{180}}\right) + 500\left(\frac{\frac{1000}{180}}{\frac{1000}{180}+\frac{100}{140}}\right)$$

On solving, $P^\circ_A = 280.5$ torr

seedha formula lgana hai bas

28. (A) This is the case of evaporation.

$P_T = P^\circ_A X_A + P^\circ_B X_B$

$= 300\ (0.6) + 800\ (0.4) = 500$ torr

(B) **Method-1** $$\frac{y_A}{1-y_A} = \frac{P^\circ_A}{P^\circ_B}\left(\frac{X_A}{1-X_A}\right)$$

$$\frac{y_A}{1-y_A} = \frac{300}{800}\left(\frac{0.6}{0.4}\right)$$

$y_A = 0.36$, $y_B = 0.64$

Method-2

$$y_A = \frac{P^\circ_A X_A}{P^\circ_A X_A + P^\circ_B(1-X_A)} = \frac{300(0.6)}{300(0.6)+800(0.4)}$$

$$y_A = \frac{180}{500} = \frac{18}{50} = 0.36$$

(C) $$\frac{y_A}{1-y_A} = \frac{P^\circ_A}{P^\circ_B}\left(\frac{X'_A}{1-X'_A}\right),\ \frac{0.6}{0.4} = \frac{300}{800}\left(\frac{X'_A}{1-X'_A}\right)$$

$X'_A = 0.8, \quad X'_B = 0.2$

(D) $P_T = P^\circ_A X'_A + P^\circ_B X'_B$

$= 300\,(0.8) + 800\,(0.2) = 240 + 160 = 400$ torr

prabhu ko yaad karo

34. This is the case of a condensation.

(1) **बेईमानी करनी है।**

Assuming $Y_A = 0.25$ when first drop of condensate formed.

$$Y_A = \frac{P^\circ_A X_A}{P^\circ_A X_A + P^\circ_B (1 - X_A)}$$

$$0.25 = \frac{300X_A}{300X_A + 800(1 - X_A)}$$

$X_A = 0.47$

(2) $P_T = P^\circ_A X_A + P^\circ_B (1 - X_A)$

$= 300 \times 0.47 + 800\,(1 - 0.47) = 565$ torr

(3) Normal boiling point is T means $P_T = 760$ torr

$760 = 300X_A + 800\,(1 - X_A)$

$X_A = 0.08$

(4) **बेईमानी करनी है।**

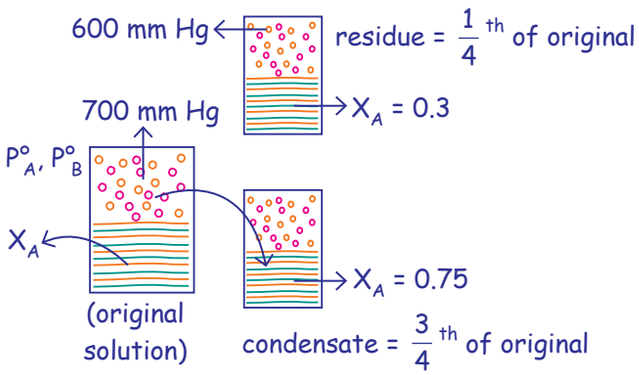
Last drop of vapour remains matlab almost sari vapour liquid phase mein aa gayi. So $X_A = 0.25$

$P_T = 300 \times 0.25 + 800 \times 0.75 = 675$ torr

(5) $Y_A = \dfrac{P^\circ_A X_A}{P_T} = \dfrac{300 \times 0.25}{675} = 0.11$

baar baar beimani karni padegi tabhi sawaal hoga

35.



$600 = P^\circ_A\,(0.3) + P^\circ_B\,(0.7)$ ...(1)

$700 = P^\circ_A\,(X_A) + P^\circ_B\,(1 - X_A)$ ...(2)

where X_A : mole fraction of A in original solution.

Let the total moles in original solution = (n)

Total moles in residue = $\dfrac{n}{4}$

Total moles in condensate = $\dfrac{3n}{4}$

moles of A in residue

$= \dfrac{n}{4} \times 0.3 = \left(\dfrac{3n}{40}\right)_{\text{moles of A in residue}}$

moles of A in condensate

$= \dfrac{3n}{4} \times \dfrac{75}{100} = \left(\dfrac{9n}{16}\right)_{\text{moles of A in condensate}}$

$$\text{Total } A = \frac{3n}{40} + \frac{9n}{16} = \frac{6n + 45n}{80} = \frac{51n}{80}$$

mole fraction of A in original solution

$$= \frac{\frac{51n}{80}}{n} = \frac{51}{80} = 0.63$$

Now solving equation (1) and (2) we get,

$P^\circ_A = 809$ mm Hg ; $P^\circ_B = 510$ mm Hg

Yeh agar kar lia, toh tum genius ho

41. Generally organic liquids are immiscible with water.

$P_T = P^\circ_A + P^\circ_B = 737$

$P^\circ_A + 707 = 737,\ P^\circ_A = 30$ torr

$$\frac{W_A}{W_B} = \frac{P^\circ_A M_A}{P^\circ_B M_B}, \quad \frac{25}{75} = \frac{30 \times M_A}{707 \times 18}$$

$M_A = 141.4$ g/mol

ganit toh aati hai na

43. (B) In the case of phenol and aniline, the intermolecular H-bonding between the phenolic proton and lone pair on the nitrogen atom of aniline is stronger as compared to respective intermolecular hydrogen bonding between similar molecules. As a result, the escaping tendency of the particles decreases in solution and hence the liquid mixture shows negative deviation.

44. (D) Negative deviation means the inter-molecular forces of attraction between the molecules of the two liquids (A-B) will be stronger than the forces present between molecules of the pure liquids (A-A and B-B).

This stronger attraction makes it difficult for molecules to escape into the vapor phase.

bhayi itna kya soch rhe ho

48. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$

9.2 mmol 10 mmol

4.2 - 5

$P_S = P^\circ_{H_2O} \times X_{H_2O}$

$$= 24 \times \left(\frac{2}{2 + 14.2 \times 10^{-3}}\right)$$

$$P_S = 24 \times \frac{2}{2 + 0.0142}$$

$$P_S = \frac{48}{2.0142} = 23.82$$

$\Delta P = 24 - 23.82 = 0.18 = 18 \times 10^{-2}$

$x = 18$

are isme equilibrium bhi agaya bhaisahab

50. Mass loss in solution $\propto P_S$

Mass gained in U tube $\propto P^\circ$

$P^\circ - P_S \propto 26.48 - 26 \Rightarrow P^\circ - P_S \propto 0.48$ g

$$\frac{P^\circ - P_S}{P_S} = \frac{n_{solute}}{n_{solvent}} \Rightarrow \frac{0.48}{26} = \frac{W_{solute}}{M_{solute}} \times \frac{M_{solvent}}{W_{solvent}}$$

$$\Rightarrow \frac{0.48}{26} = \frac{20}{M_{solute}} \times \frac{18}{250} \quad (\because d_{water} = 1 g/mol)$$

$$M_{solute} = \frac{26 \times 20 \times 18}{0.48 \times 250} = 78 g/mole$$

53. First calculate ΔT_b as:

$\Delta T_b = (100 + 273) - (99.63 + 273) = 0.37$ K

Now, calculate ΔT_b as:

$$\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1}$$

$$\Rightarrow w_2 = \frac{\Delta T_b \times M_2 \times w_1}{K_b \times 1000} = \frac{0.37 \times 342 \times 500}{0.52 \times 1000}$$

= 121.67 g (approximately)

calculate toh kar hi loge na

54. (B) 8% aqueous solution means 8 g of solute is present in 100 g of solution.

According to the given information,

$(\Delta T_b)_A = (\Delta T_b)_B$

$$\cancel{K_b}\left(\frac{\frac{2}{M_A}}{\frac{98}{\cancel{1000}}}\right) = \cancel{K_b}\left(\frac{\frac{8}{M_B}}{\frac{92}{\cancel{1000}}}\right)$$

$$\frac{2}{MA} = \frac{8}{MB}, 4 M_A = M_B$$

Calculate toh kar hi loge na?

58. According to the given information:

$$\frac{(T_B)_x}{(T_B)_y} = \frac{2}{1}, \left(\frac{(\Delta H)_x}{(\Delta H)_y}\right) = \frac{1}{2}$$

Also, $\Delta T_b = K_b m$

$$\frac{(\Delta T_B)_x}{(\Delta T_B)_y} = m = \frac{(K_B)_x \times \text{molality}}{(K_B)_y \times \text{molality}}$$

$$= \frac{(T.B)_x^2}{(T.B)_y^2} \times \frac{\Delta H_y}{(\Delta H)_x} = (2)^2 \times 2 = 8$$

darr toh nahi gaye

62. (A) ΔT_f is maximum when (i × m) value is maximum.

Also, molality (m) is maximum for compound with lesser molar mass of solute.

Among the given compounds, acetic acid has lowest molar mass.

For option (A),

$$m = \frac{\frac{180}{60}}{M_{solvent}(kg)} \text{ and } i = 1 + a$$

For option (B),

$$m = \frac{\frac{180}{60}}{M_{solvent}(kg)} = \text{ and } i = 0.5$$

Therefore, (i × m) value for option (A) is greater than that of option (B). Hence, ΔT_f will be maximum for option (A).

Are asaan hai, mushkil nahi hai

63. The formula to calculate depression in freezing point is:

$\Delta T_f = K_f m$

m = Moles of solute/ mass of solvent (in kg)

Let w be the mass of solute.

The above expression in the terms of mass of solute:

w (mass of solute) = $\frac{m \times \Delta T_f \times W}{1000 \times K_f}$

Substitute the given values in the above expression:

$= \frac{60 \times 0.186 \times 500}{1000 \times 1.86} = 3g$

bas formula hi toh lagaya hai

67. Molality of solution of 0.75 m which means that 0.75 moles of sucrose present in 1 kg of solvent.

$\therefore m_{solution} = m_{water} + m_{sucrose} = 1000 + .75 \times 342 =$ 1256.5 gm

Calculate molality of solution as:

$\therefore m_{solution}$ (in kg) $= \frac{256.5}{1256.5} \times 1000 = 204.14$ gm, .

$m_{water} = 1000 - 204.14 = 795.86$ gm

Finally, calculate depression in freezing point as:

$\therefore \Delta T_F = K_F \times m \Rightarrow 4$

$= 1.86 \times \frac{204.14/342}{W/1000} \Rightarrow w = 277.15$

Amount of ice = 795.86 – 277.15 = 518.31 gm

(thoda toh dimag ka istemaal karo)

69. The formula to calculate depression in freezing point is:

$\Delta T_f = K_f\, m$

m= Moles of solute/ mass of solvent in kg

$\Delta T_f = \frac{3.9 \times 10.2 \times 1000}{176 \times 150}$

On solving, $\Delta T_f = 1.506 = 15.06 \times 10^{-1}$

72. Osmotic pressure=CRT

$1.5 = C \times 0.0821 \times 300$, $C = 0.0609$ M

74.

$$HX \rightleftharpoons H^+ + X^- \qquad K_a = 1.2 \times 10^{-5}$$

At t = 0 0.03M

At t = eq 0.03–x x x

$K_a = 1.2 \times 10^{-5} = \frac{[H^+][X^-]}{[HX]} = \frac{x^2}{0.03 - x}$

$0.03 - x \approx 0.03$ ($\because K_a$ is very small)

$\frac{x^2}{0.03} = 1.2 \times 10^{-5}$

Hence, $x = 6 \times 10^{-4}$

Final solution : $0.03 - x + x + x$

$0.03 + x = 0.03 + 6 \times 10^{-4}$

Osmotic pressure, $\pi = C \times R \times T$

Given: T = 300 K

$(0.03 + (6 \times 10^{-4}) \times 0.083 \times 300$

$= 76.19 \times 10^{-2} \approx 76 \times 10^{-2}$

(prabhu ke gun gaao)

78. First calculate the moles of glucose and urea:

Moles of glucose = $\frac{\text{Mass}}{\text{Molar mass}} = \frac{9}{180}$ = 0.05 mol

Moles of urea = $\frac{12}{60}$ = 0.2 mol

Total moles = 0.05 + 0.2 = 0.25 mol

Calculate concentration as:

Concentration = $\frac{n}{V} = \frac{0.25}{1}$ = 0.25 M

Temperature = 27 + 273 = 300 K

Osmotic pressure = CRT = 0.25 × 0.08206 × 300 = 6.155 atm

(gazab hai yeh sawaal)

81. (A) KNO_3 is a strong electrolyte and undergoes complete dissociation and its van't Hoff factor is equal to 2.

For CH_3COOH, the van't Hoff factor is less than KNO_3 because it is a weak electrolyte. So, according to this,

$i_1 > i_2$, $P_1 > P_2$

ab' i' bhi agaya sawaalo mei

83. (B) $\Delta T_F = (i\, K_F)m$

$= [1 + (n - 1)\alpha]\, K_F.m$

For (A), i = [1 + (2 - 1)0.9] = 1.9

$\Delta T_f = iK_f m = 0.45$ m

For (B), i = [1 + (3 - 1)0.7] = 2.4

$\Delta T_f = iK_f m = 0.72$ m

For (C), i = [1 + (3 - 1)0.7] = 2.4

$\Delta T_f = iK_f m = 0.48$ m

For (D), i = [1 + (1 - 1)0] = 1

$\Delta T_f = iK_f m = 0.4$ m

Hence, for B, depression in freezing point is maximum.

84. (B) Calculate Van't Hoff Factor (i) for AX_2

$\Delta T_b = i \times K_b \times m$

$0.0156 = i \times 0.52 \times 0.01$, $i = 3$

Since $i = 3$, AX_2 fully ionises

Calculate Van't Hoff Factor (i) for AY_2

$\Delta T_b = i \times K_b \times m$

$0.0260 = i \times 0.52 \times 0.0508$

$$i = \frac{0.0260}{0.0264} = \approx 1$$

Since $i \approx 1$, AY_2 does not ionise (remains unionised).

87. $\Delta T_f = 0.06 = iK_F m$

$0.06 = i \times 1.86 \times 0.025$

$i = 1.29$

$i = 1 + (n - 1)\alpha$

$1.29 = 1 + (2 - 1)\alpha$

$\alpha = 0.29$

Now,

$$HA_{(aq)} \rightleftharpoons A^-_{(aq)} + H^+_{(aq)}$$

C	0	0
$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$K_\alpha = \frac{C\alpha^2}{1-\alpha} \simeq C\alpha^2$$

$= 0.025(0.29)^2 = 2.1 \times 10^{-3}$

yeh kya equilibrium phirse agaya

90. The amount of solvent is calculated as,

$= 100 - (29.25 + 19) = 51.75$ g

Now, the elevation in boiling point is calculated as,

$$\Delta T_b = \left[\frac{2 \times 29.25 \times 1000}{58.5 \times 51.75} + \frac{3 \times 19 \times 1000}{95 \times 51.75}\right] \times 0.52$$

$\Delta T_b = 16.075$

The boiling point of sea water is the normal boiling point of water plus the change in boiling point.

Therefore, boiling point of sea water is,

$(T_b)_{solution} = 100 + 16.07 = 116.07°C$

92. Use Henry's law: $P_{gas} = K_H.X_{gas}$

Substitute values in above equation:

$$0.835 = 1.67 \times 10^3 \times \frac{n(CO_2)}{\frac{0.9 \times 1000}{18}}$$

$n(CO_2) = 0.025$

Millimoles of $CO_2 = 0.025 \times 1000 = 25$ mmol

98. (C) (A) $K_H \propto \frac{1}{\text{solubility}}$

α has lowest solubility.

(B) on ↑ing temperature, solubility of gases ↓ses.

(C) $P_\delta = (k_H)_\delta \times \chi_\delta$

$$= 0.5 \times \frac{55.5}{55.5 + \frac{1000}{18}} = \frac{0.5}{2}$$

$= 0.249 = 249$ bar

(D) $P_\tau = K_H \times \chi_\tau$

$$= 2 \times 10^{-5} \times \frac{55.5}{55.5 + 1000/18}$$

$= 10^{-5}$ kbar $= 10^{-2}$ bar

dimag ki ghanti baji?